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2026-09

Introduction to Open Research Data

Why should we share data? (And do I get a co-authorship?)

Reema Gupta, Ludwig-Maximilians-Universität München using material from Prof. Dr. Felix Schönbrodt

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Why open data?

Why open data?

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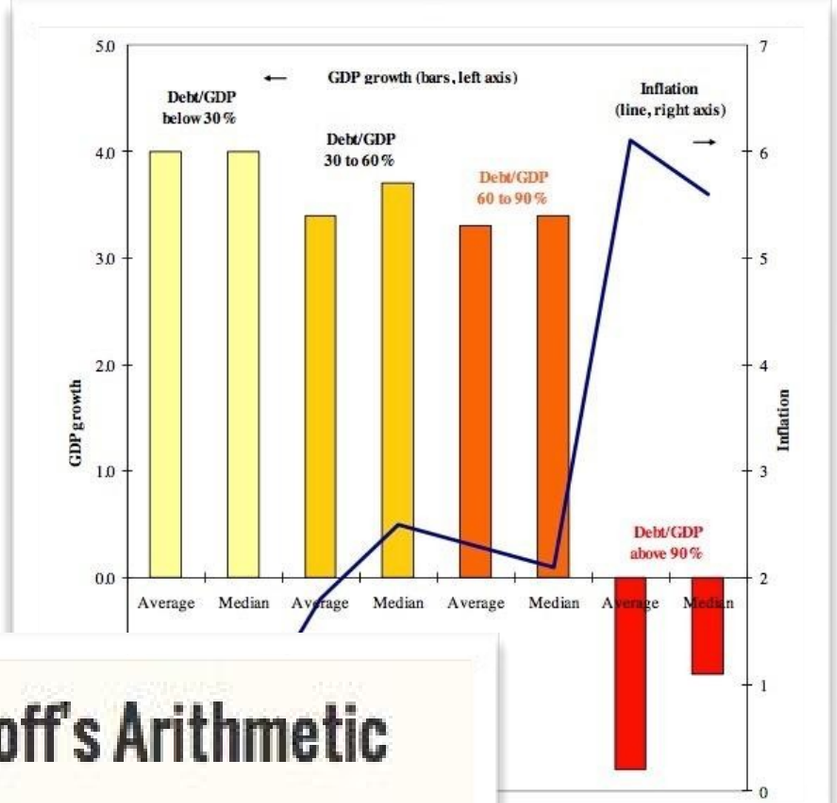
1. *Nullius in verba* -
take nobody's word for it

- Motto of the oldest scientific society (Royal Society, founded 1660)
- Science is not built upon blind trust, but on verifiability.
- „Organized skepticism“ (Merton, 1942)



Only when raw data (and other research material) are shared organized skepticism can be enacted, and science can really be self-correcting. Open data is one part of good scientific practice.

90%-Excel-Gate



How Much Unemployment Was Caused by Reinhart and Rogoff's Arithmetic Mistake?

Published: 16 April 2013

The Excel Depression

APRIL 18, 2013



Paul Krugman

In this age of information, math errors can lead to disaster. NASA's Mars Orbiter crashed because engineers forgot to convert to metric measurements; JPMorgan Chase's "London Whale" venture went bad in part because modelers divided by a sum instead of an average. So, did an Excel coding error destroy the economies of the Western world?

The most important point of the story: The original authors shared their raw data, which made it possible to correct the honest mistake!

http://economix.blogs.nytimes.com/2013/04/16/flaws-are-cited-in-a-landmark-study-on-debt-and-growth/?smid=tw-share&_r=0

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2. *Efficiency and Inclusiveness*

- Speedy responses in outbreaks; share rare and hard-to-collect data

OECD Policy Responses to Coronavirus (COVID-19)

Why open science is critical to combatting COVID-19

Updated 12 May 2020

Matt Might
@mattmight

Let's remember this after we cure COVID19. Let's remember how open science shortened the time to find a treatment. And, let's never go back to the old ways again. The moral necessity of open access to all taxpayer-funded research is unassailable now.

Tweet übersetzen

The Economist @TheEconomist · 17. März 2020

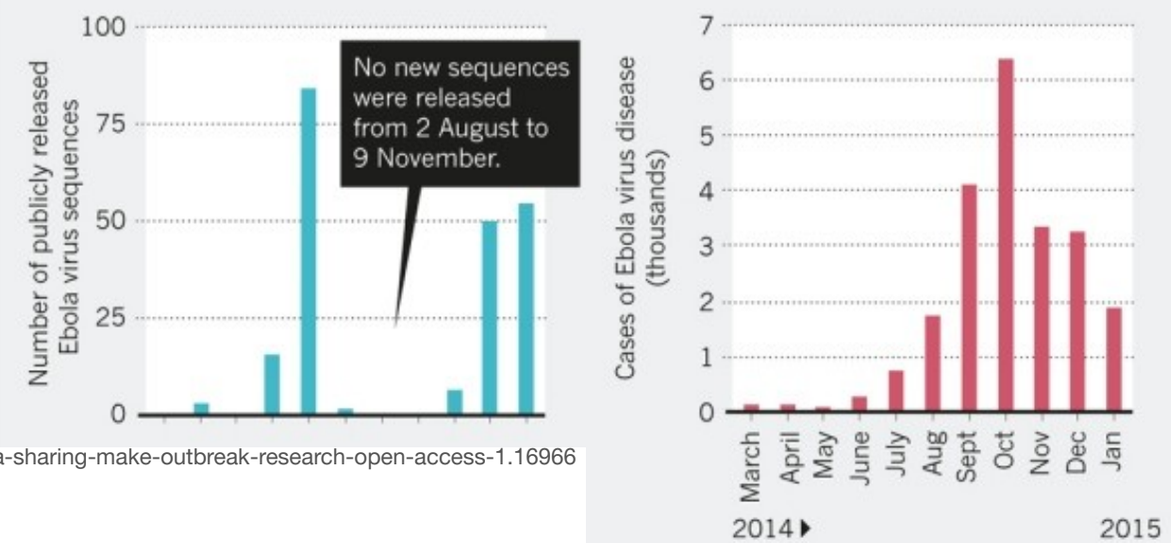
There is no treatment for 90% of rare diseases. Personalised medicines can target such illnesses econ.st/3cZVmN6

Data sharing: Make outbreak research open access

Nathan L. Yozwiak, Stephen F. Schaffner & Pardis C. Sabeti

GAPS IN THE DATA

Genome sequences from the West Africa outbreak of Ebola virus were first made publicly available in April 2014. Since 99 genomes were released in July, data sets have been shared sporadically, even though more are known to have been generated.



https://www.nature.com/news/data-sharing-make-outbreak-research-open-access-1.16966

The covid-19 pandemic has shown how fast scientific progress can be when we share our data and knowledge freely, and that free knowledge is a moral imperative.

Why open data?

3. Public money = public good



https://publiccode.eu

Publicly funded research data does not belong to the researcher who collected it. They have the right of primary usage, but after that the data should be considered a public good (of course respecting privacy rights and applicable copyrights).

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4. Data persistence

- never lose data due to a crashed hard disk drive
- Open data allows independent copies at multiple places (e.g. even in decentralized bittorrent swarms) – create resilience

against autocratic regimes that delete „inconvenient“ research data



A publicly funded repository is the right place for long term storage of research data – not your private USB stick, your personal university website (that vanishes after you change affiliation), or the journal’s online supplemental material that hides the data behind a paywall. Prefer repositories in countries that value academic freedom.

Availability Prevalence

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Data & analysis script availability (prevalence estimates)

	Data	Analysis scripts
Psychology (2014-2017)[1]	2% [1-4%]*	1% [0-1%]
Social Sciences (2014-2017)[2]	7% [2-13%]	1% [0-3%]

[1]Hardwicke et al. (2021)

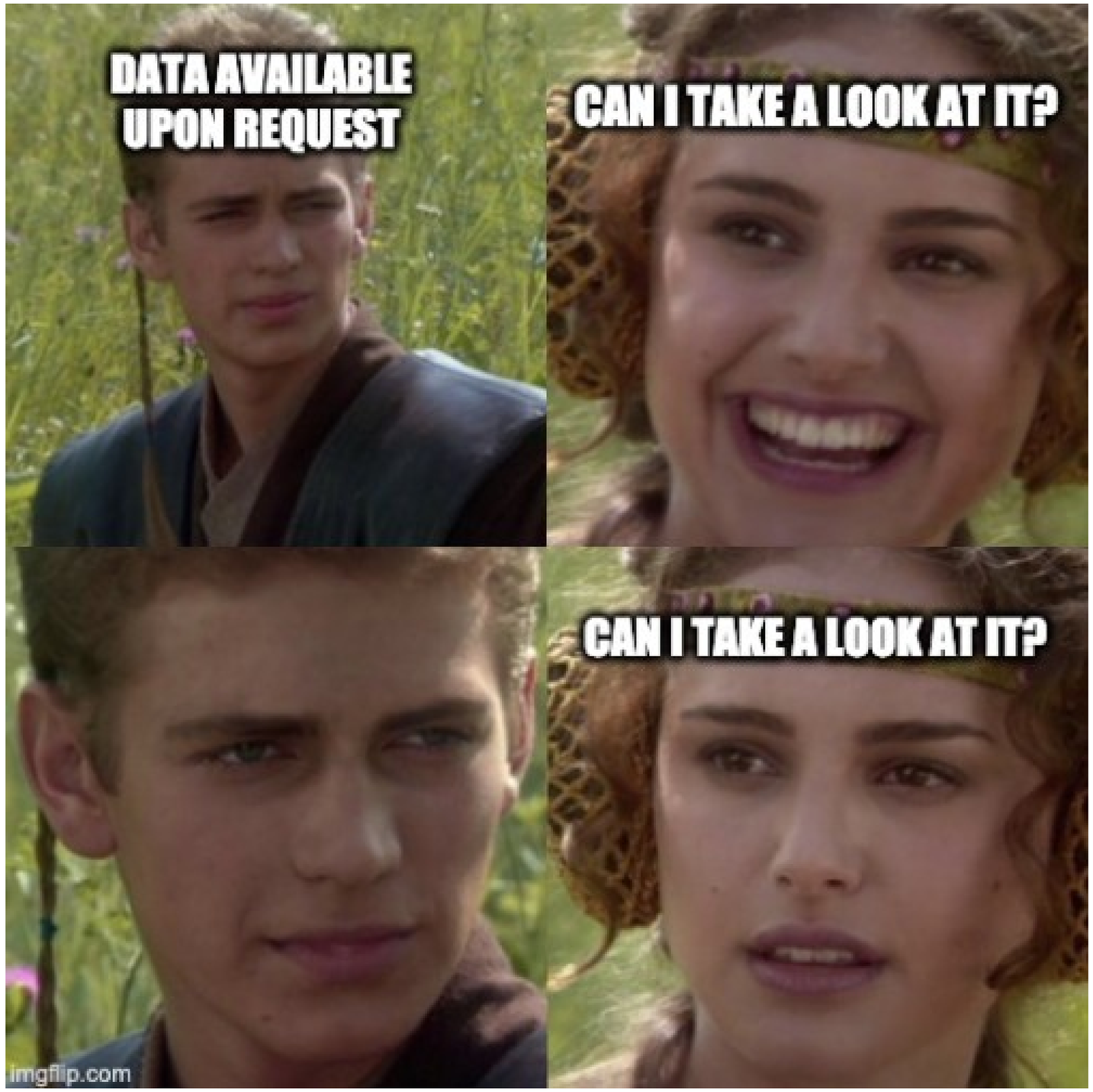
[2]Hardwicke et al. (2020)

*[95% confidence intervals]

Hardwicke et al. (2024): random samples of psychology articles

Data availability 2% → 14%; code availability 1% → 8.5%

Hardwicke et al. (2022), Perspectives on Psychological Science 17(1): 239–251; Hardwicke et al. (2024), Advances in Methods and Practices in Psychological Science 7(4).



https://twitter.com/JohnBorghi/status/1405227318876905472?s=20

„Available upon request“

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Data availability on request (selected studies)

	Data shared
141 articles published in four major APA journals (2004)[3]	27%
516 ecology articles published (1991-2011)[4]	20%
111 most highly-cited psychology & psychiatry articles (2006-2016)[5]	14%

[3]Wicherts et al. (2006)
[4]Vines et al. (2014)
[5]Hardwicke & Ioannidis (2018)

Gabelica et al. (2022): wrote to authors of 1,792 articles whose data availability statement promised sharing on request

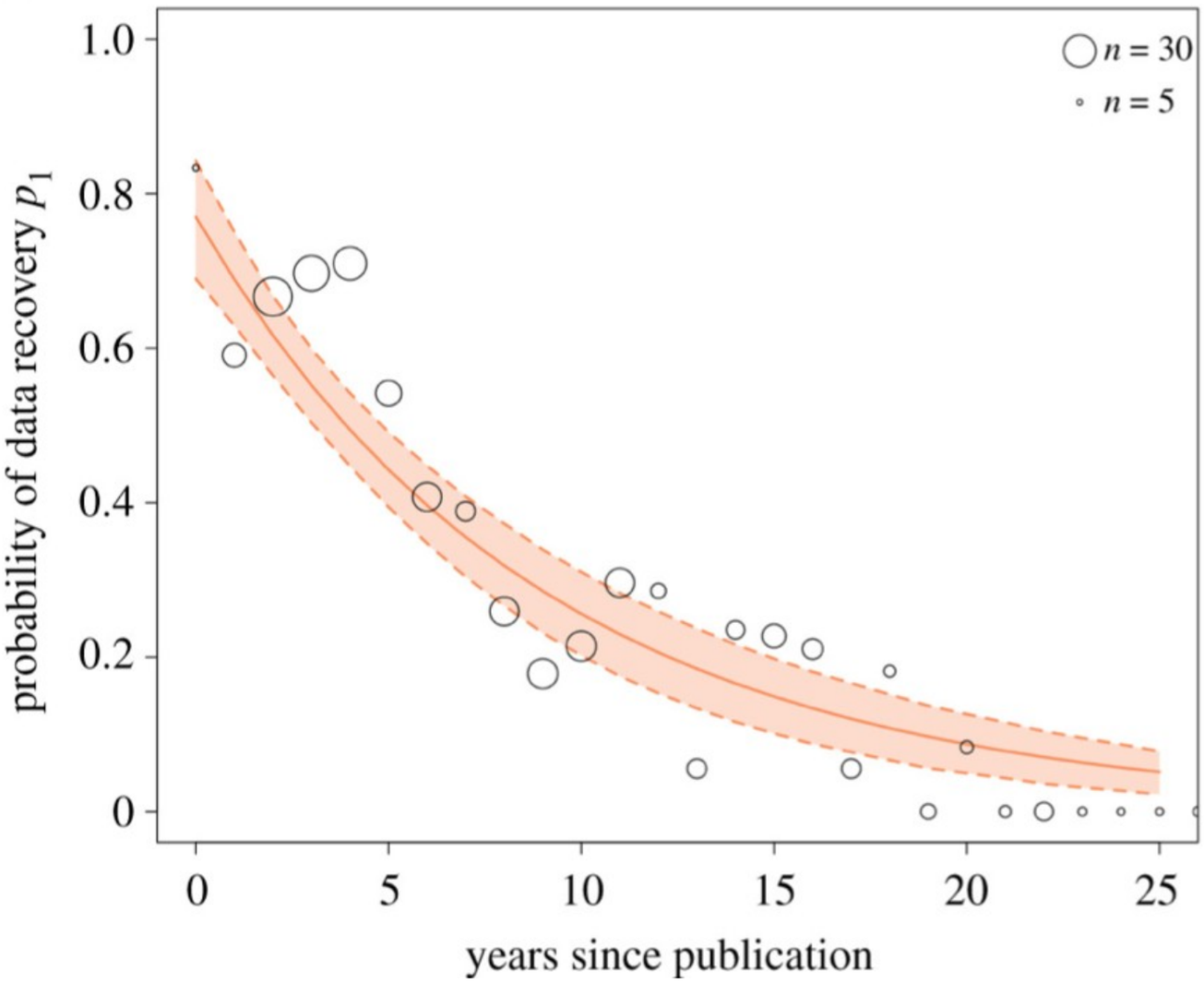
93% never responded or declined. 6.8% shared data enabling reanalysis

Medicine (Hamilton et al. 2023): ~8% of articles declare data availability, ~2% actually share it

Tedersoo et al. (2021), 875 Nature & Science papers, nine disciplines: „upon request“ delivered 42.3%, no statement at all 38.3%.

Gabelica, Bojčić & Puljak (2022), Journal of Clinical Epidemiology 150: 33–41; Tedersoo et al. (2021), Scientific Data 8: 192.

Data decay in social learning research



- „probability of finding material for any publication halves every 5.7 years“
- „probability of recovering data for studies > 20 years ago is close to zero“

Across all fields: the odds a dataset still exists fall ~17% per year (Vines et al. 2014)

Minocher, R., Atmaca, S., Bavero, C., McElreath, R., & Beheim, B. (2021). Estimating the reproducibility of social learning research published between 1955 and 2018. Royal Society Open Science, 8(9), 210450. https://doi.org/10.1098/rsos.210450 | Vines, T. H., et al. (2014). The availability of research data declines rapidly with article age. Current Biology, 24(1), 94-97.

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5. More and more funders and journals demand it.

Journal	Total	Data Citation	Data Transparency	Analysis Code Transparency	Materials Transparency
The Journal of Politics University of Chicago Press	9	0	3	3	0
Journal of Economic Perspectives American Economics Association	9	2	3		
Journal of Economic Literature American Economics Association	7	0	3		
American Journal of Political Science Wiley	11	0	3		
Nature Methods Springer Nature	11	1	3	1	1
AEA Papers & Proceedings American Economics Association	7	0	3	3	0
Journal of Legal Studies University of Chicago Press	10	1	3	3	3
Journal of Experimental Political Science Cambridge University Press	16	1	3	3	3
Journal of Peace Research SAGE	10	1	3	3	3
Biometrical Journal Wiley	7	0	3	3	0

TOP GUIDELINES
TRANSPARENCY AND OPENNESS PROMOTION

https://www.topfactor.org

TOP level 3: Data must be posted to a trusted repository, and reported analyses will be reproduced independently prior to publication.

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5. More and more funders, journals and policies demand it.

DFG Code of Conduct (2019): data availability is part of good scientific practice, binding for German universities; DFG (2022): requires statement on research data handling and allows to request the associated costs

Horizon Europe (2021): data management plan + trusted repository are mandatory, „as open as possible, as closed as necessary“

NIH (2023): a data management & sharing plan for every grant, at the world's largest public research funder

Not a values debate anymore. A funding requirement.

DFG (2019) Guidelines for Safeguarding Good Research Practice; Horizon Europe AGA; NIH Data Management & Sharing Policy (2023).

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What is open data?

What is Data?

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"The recorded factual material commonly retained by and accepted in the scientific community as necessary to validate research findings."

(EPSRC, 2018)

Anything and everything produced in the course of research.

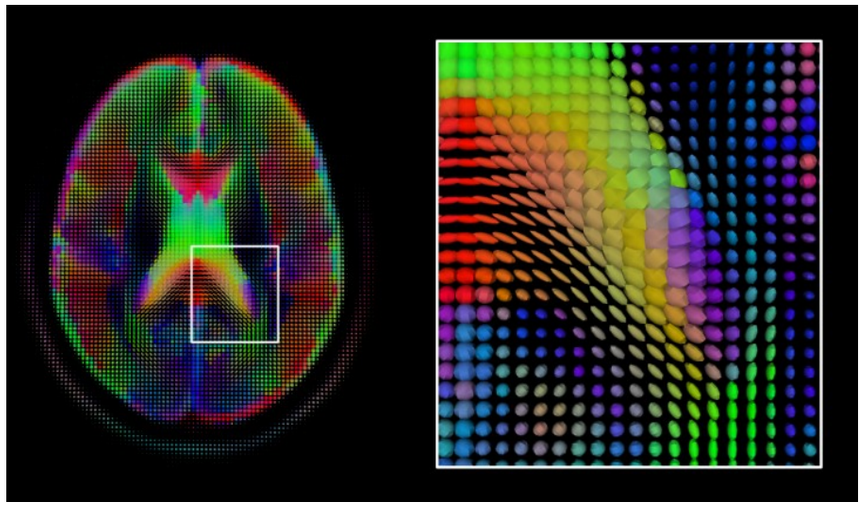
Lynch (2014)

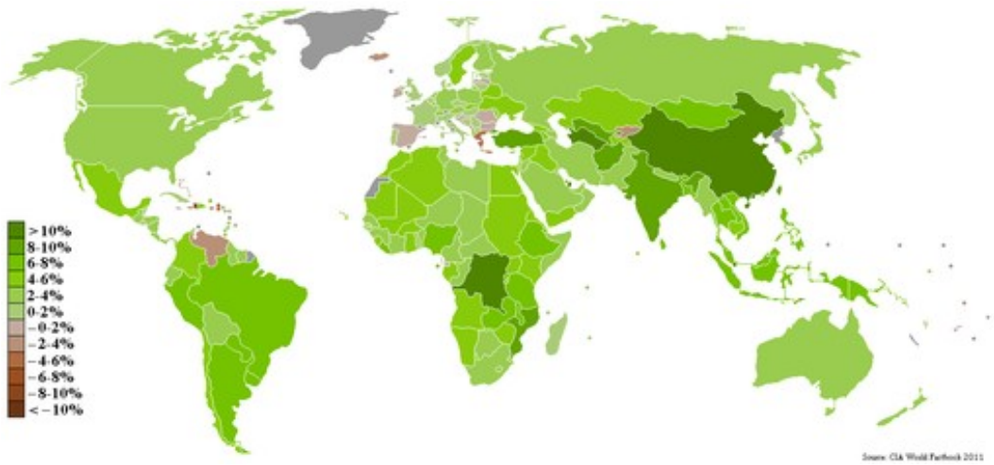


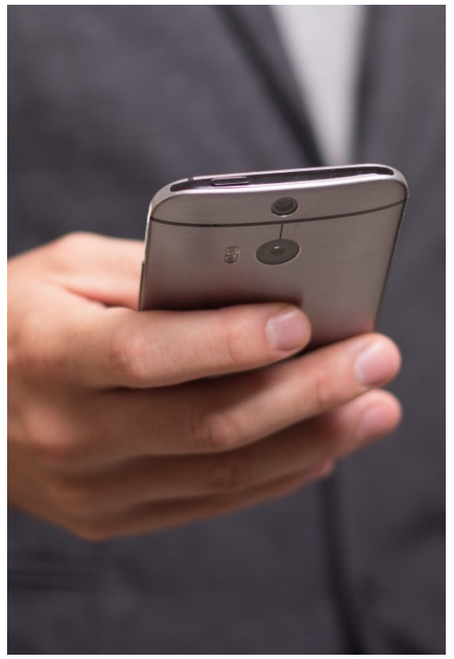
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What is Data?

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→ we need field-specific definitions: What constitutes „research data“?

Pictures from en.wikipedia.org/w/index.php?curid=36808161; pixabay.com; pxhere.com/en/photo/1192476; pxhere.com/en/photo/914805; pxhere.com/en/photo/595225; commons.wikimedia.org/wiki/File:Gdp_real_growth_rate_2007_CIA_Factbook.PNG

Example: Psychology

Thus, primary data in psychology are unaltered (i.e., untransformed, not aggregated, etc.) quantitative or qualitative data available in digital form, e.g.:

- Each manipulated and measured variable of every experimental session of every study participant in an experiment;
- Each response of every person to each item in a survey;
- Original wording of inputs in free text fields;
- Digitized video recordings;
- [...].

Primary data also include the data of cases that were excluded from the analyses (with the exception of those cases in which participants withdrew their consent during or after data collection).
To summarize, we define primary data as the set of all data points collected during the course of a study or project, as initially digitized, but otherwise in a completely unaltered form.

Recommendations of the German Psychological Society, https://psyarxiv.com/24ncs/

DFG Subject Specific Recommendations

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Fachspezifische Empfehlungen zum Umgang mit Forschungsdaten

- Stellungnahme der Akademie für Soziologie zum Umgang mit Forschungsdaten in der Soziologie (PDF | 219 KB)
- Stellungnahme des Fachkollegiums 111 „Sozialwissenschaften" zum Forschungsdatenmanagement in der Soziologie, der Politikwissenschaft und der Kommunikationswissenschaft (PDF | 100 KB)
- Stellungnahme zum Umgang mit Forschungsdaten in der Erziehungswissenschaft, Bildungsforschung und Fachdidaktik (PDF | 803 KB)
- Handreichung des Fachkollegiums 101 zum Umgang mit Forschungsdaten (PDF | 45 KB)
- Handreichung des Fachkollegiums 106 zum Umgang mit Forschungsdaten (PDF | 124 KB)
- Bereitstellung und Nachnutzung von Forschungsdaten in der Soziologie: Stellungnahme des Vorstands und Konzils der DGS (PDF | 761 KB)
- Digitaler Wandel in der Wissenschaft: Herausforderungen und Chancen für das Fachgebiet Materialwissenschaft und Werkstofftechnik (PDF | 92 KB)
- Information für die Wissenschaft Nr. 66/2015: DFG verabschiedet Leitlinien zum Umgang mit Forschungsdaten ›
- Information für die Wissenschaft Nr. 36/2015: Richtlinien zum Umgang mit Forschungsdaten in der Biodiversitätsforschung ›
- Richtlinien zum Umgang mit Forschungsdaten in der Biodiversitätsforschung (PDF | 192 KB)

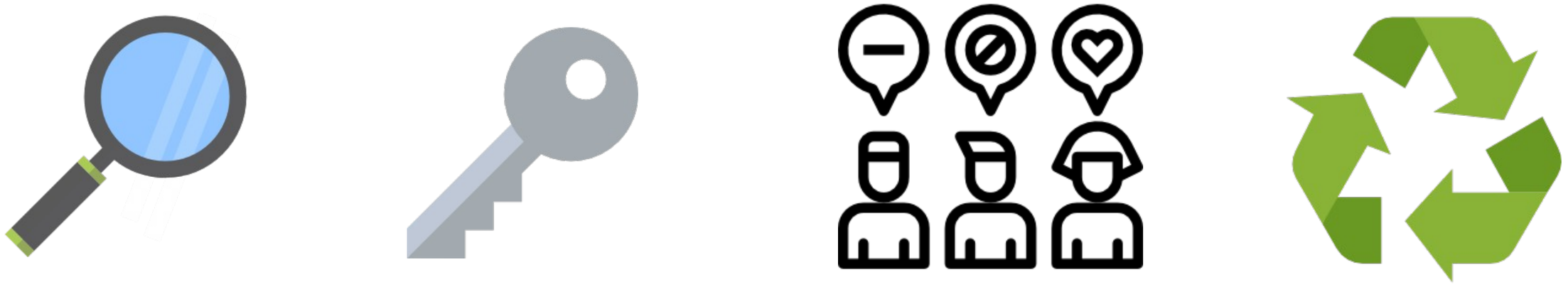
https://www.dfg.de/foerderung/antrag_gutachter_gremien/antragstellende/nachnutzung_forschungsdaten/index.html

https://psyarxiv.com/

Not only open, but FAIR

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The FAIR principles



Findable **A**ccessible **I**nteroperable **R**eusable

Good research data management

→ *attend workshop on*
„FAIR research data management“

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FAIR enough aka. FAIR for dummies

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Findable:

- ✓ (1) Put your data in a professional repository that provides a globally unique, citable and persistent identifier (e.g., DOI, PURL, ARK) and requires essential metadata such as a standardized codebook (e.g., PsychArchives, NFDI, see also https://www.re3data.org/)



Accessible:

- ✓ This is automatically achieved with (1)



Inter-operable:

- ✓ Most interoperability is automatically achieved with (1)
- ✓ (2) Link to other related data sets (optionally, if applicable). This is a feature of the repository.



Reusable:

- ✓ (3) Explicitly assign a license, as open as possible (e.g. CC0; see OSC tutorial on licenses)
- ✓ (4) Describe (in the metadata) people, entities, and processes involved in the creation, management and longer-term curation of the data set: (a) Data collection date; (b) Contributors involved in data creation and their roles (DataCRediT); (c) Data versioning information (if applicable)
- ✓ (5) Write a structured codebook (following the standards of your field)
- ✓ (6) Use a structured, open, non-proprietary format for the data (e.g., .csv, .tab, .json, .xlsx; **not** .sav or .pdf)

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Balancing values: Three fields of tension with data

Balancing values I

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Open Data, public interest/entitlement to publicly funded data



Privacy rights of research subjects

- Privacy rights > openness; but GDPR also mentions a „legitimate interest“ of research
- Ask participants for a broad consent of open reuse
- Use tiered access levels (e.g., „scientific use files“)
- Publish aggregated data (e.g., ratings of videos) without the primary data (videos)
- Sharing something > sharing nothing
- It’s not either „open“ or „closed“ - it’s a continuum:
  As open as possible, as restricted as necessary

Balancing values II

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Right of first usage, incentives to collect data in the first place

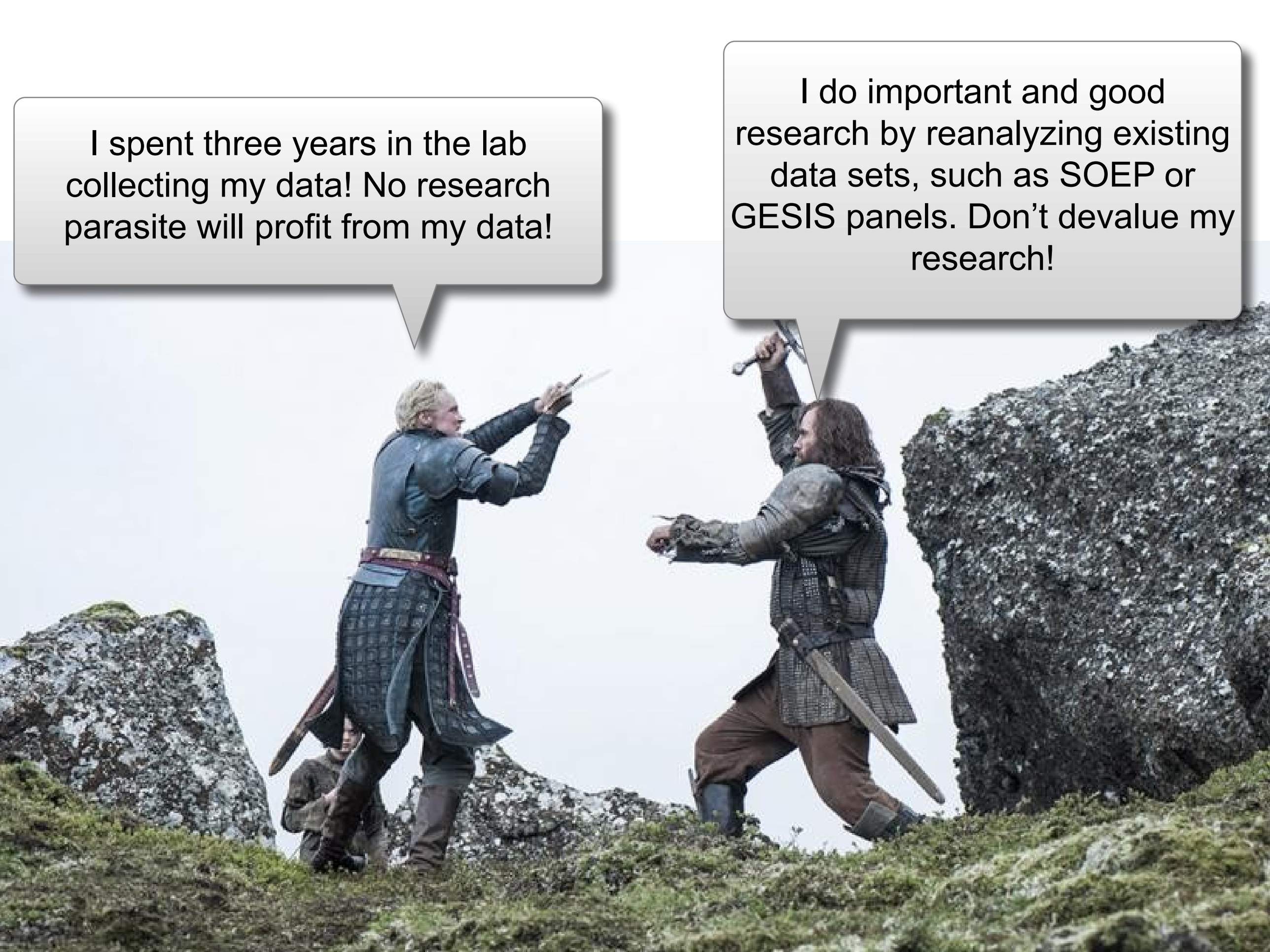


Optimal and efficient gain of knowledge by data reuse

- Incentivize data sharing
- Right of first usage, possibility of embargo
- At the end of the day (resp., the embargo), all data are as open as possible

I spent three years in the lab collecting my data! No research parasite will profit from my data!

I do important and good research by reanalyzing existing data sets, such as SOEP or GESIS panels. Don’t devalue my research!



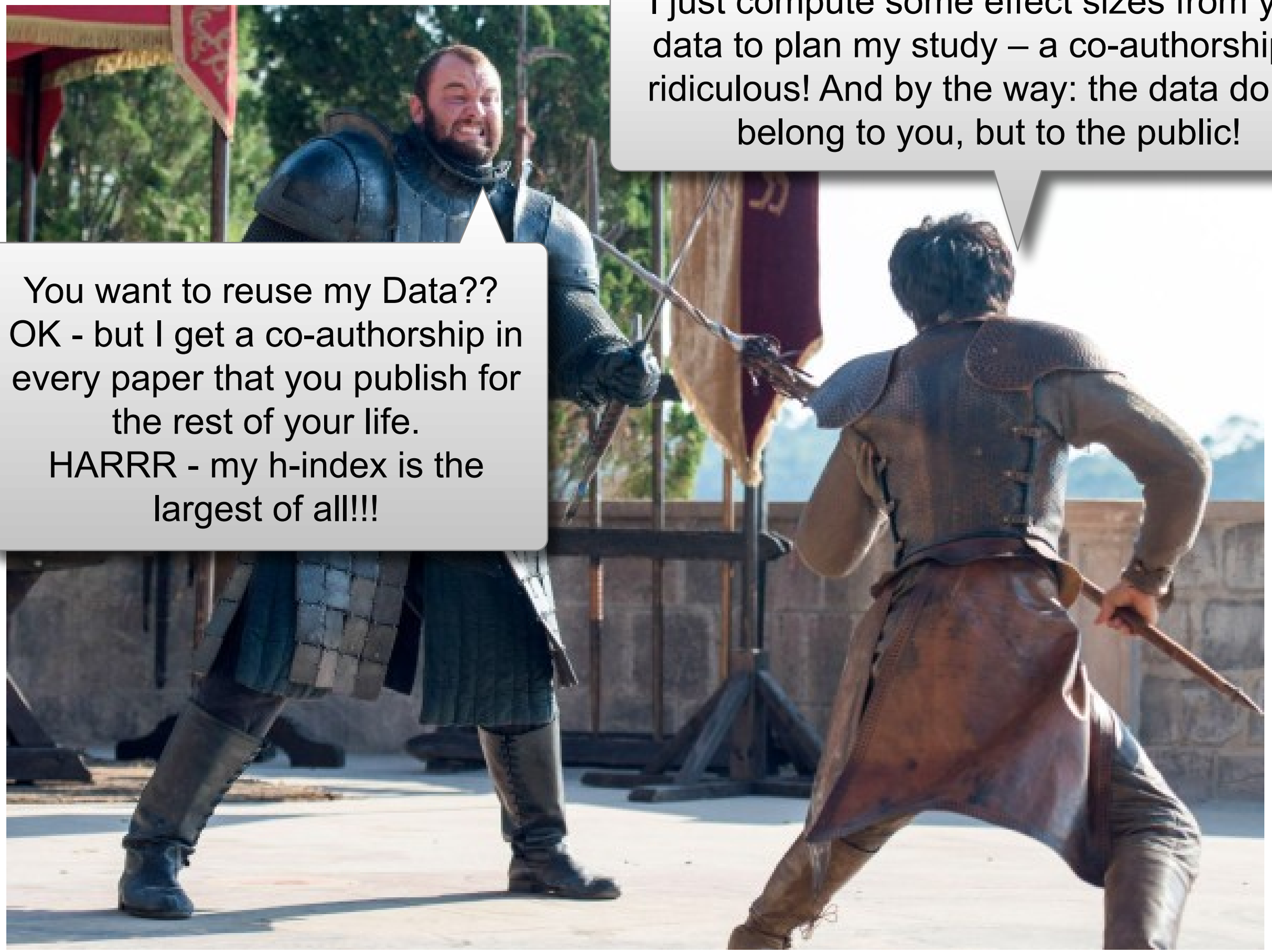
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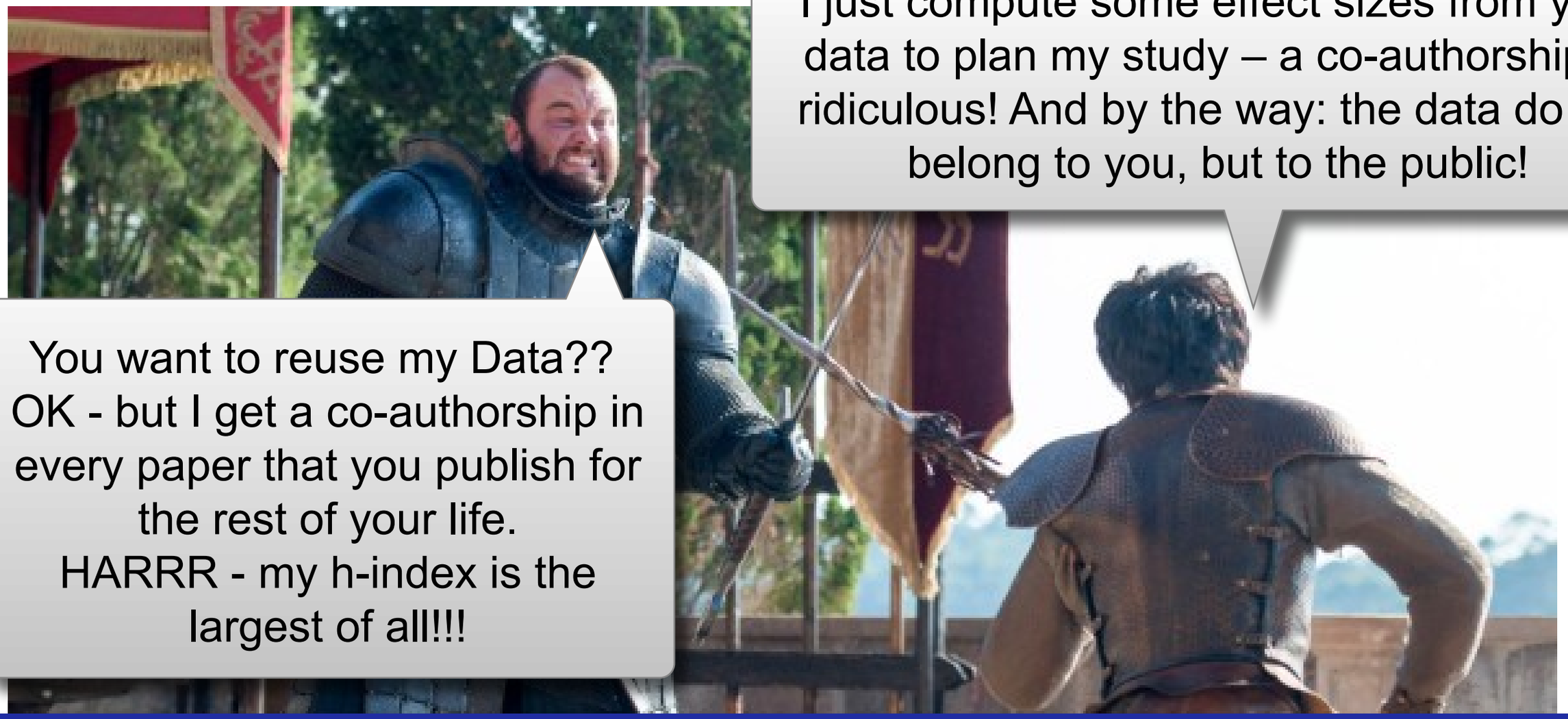
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A proposed answer:

- Collecting primary data is a sine qua non for empirical research, so this should be valued.
- But: No type of research is inherently „better“, there is a place for researchers purely focusing on re-analysis.
- Job committees should define what kind of research expertise they are looking for - some might prefer persons with data collection experience, some might look for data re-analysts.


I just compute some effect sizes from your data to plan my study – a co-authorship is ridiculous! And by the way: the data do not belong to you, but to the public!
You want to reuse my Data?? OK - but I get a co-authorship in every paper that you publish for the rest of your life. HARRR - my h-index is the largest of all!!!



A proposed answer:

- Co-authorship can already difficult in single-lab papers - cannot be solved in an abstract way, rather single case decisions
- „Simple reuse“ does not imply co-authorship, but there are cases where an offering is warranted
- Co-authorship requires something more than only providing already existing data
- Move from „authorship“ to „contributorship“ (CRediT)

Balancing values III

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Reproducibility and verifiability of published analyses



Protect original authors against inadequate burden and potential attacks

- Primary focus: openness and transparency. Correcting errors is painful, but a necessary condition for doing science
- Data providers should be informed if their data are going to be reused or reanalyzed → allows to prepare a reaction

Balancing values III

Reproducibility and verifiability of published analyses



Protect original authors against inadequate burden and potential attacks

- Problematic asymmetry:
  - Data provided → often errors get detected
  - No data provided → no errors are detected (because not possible). Default assumption: „Everything is OK. Perfect paper, because no errors are spotted!“
- Making oneself vulnerable is good for science, and should also be good for reputation!
- Change default assumption?
  „No data → Probably erroneous analysis.“

Nuijten et al. (2016), Behavior Research Methods 48: ~50% of psychology articles contain at least one inconsistently reported statistic.

Incentivizing Open Data Sharing

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State of Open Data 2025 (4,700+ researchers, 151 countries): **81% support open data as common scholarly practice**

Familiarity with FAIR principles: 15% (2018) → 41% (2025)

But 69% say researchers get too little credit for sharing data (2020: 78%)

CoARA

„6. Review and develop research assessment criteria, tools and processes:

[…] The changes should increase the ability to assess quality by enabling recognition of all contributions to quality research by research units and institutions. Such recognition includes that of **early sharing of data** and results, open collaboration, teamwork […]“

DFG Deutsche Forschungsgemeinschaft

„Performance assessment based on content-related qualitative criteria also explicitly includes ensuring that the **entire spectrum** of academic publication types are equally displayed and acknowledged in funding proposals and CVs“

„including articles on preprint servers, **data sets** or software packages, for example“

https://doi.org/10.6084/m9.figshare.30823079; https://coara.eu/agreement/the-commitments/; DFG Information für die Wissenschaft Nr. 61 (2022): Maßnahmenpaket zum Wandel der wissenschaftlichen Bewertungskultur

Incentivizing Open Data Sharing

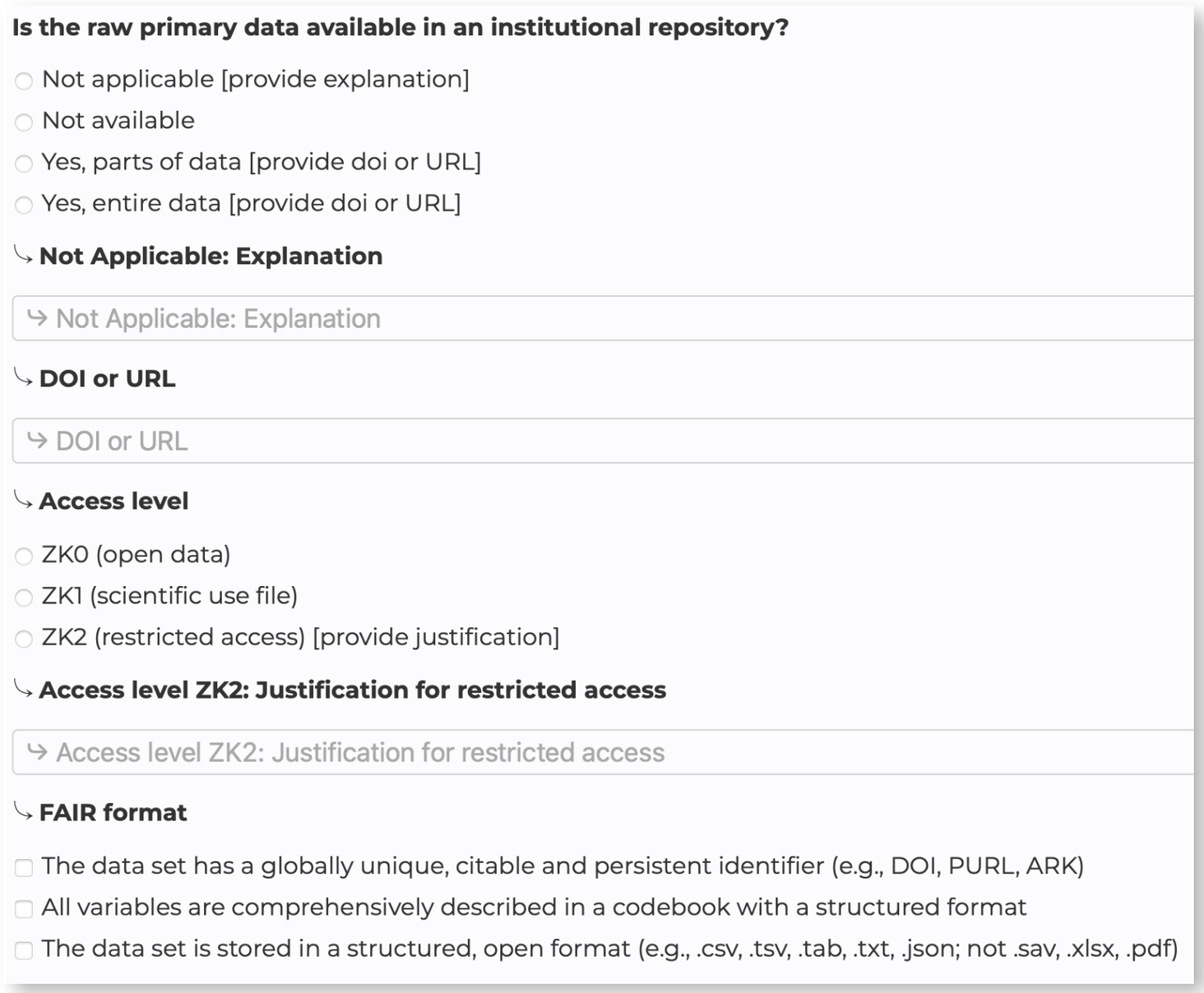
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RESQUE (Research Quality Evaluation Framework)

Principle 1: Academic contributions are multifaceted. Regarding research contributions, do not only value (a) journal articles, but also (b) data sets and (c) research software development.

Schönbrodt et al. (2024), Responsible Research Assessment I: Implementing DORA and CoARA for hiring and promotion in psychology

(Working commissioned by the German Psychological Society)



https://www.resque.info/

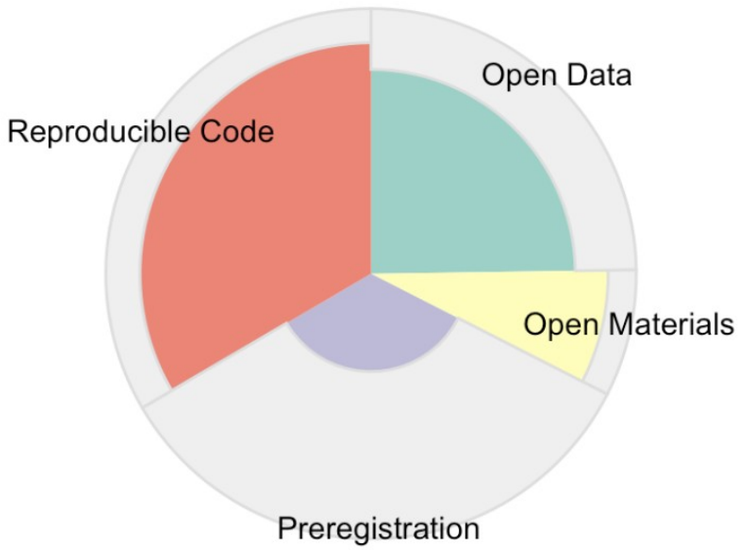
Incentivizing Open Data Sharing

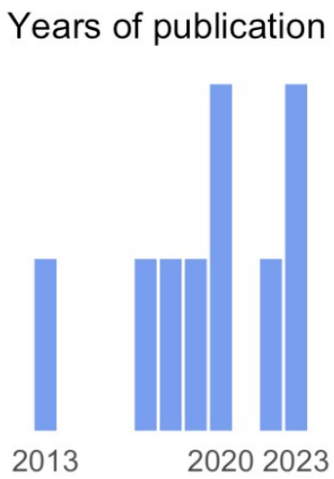
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RESQUE Profile for job candidates:

Relative Rigor Score	Open Data	Open Material	Preregistration	Reproducible Code
50%				

Relative Rigor Score





Overall score = 50%. Chart is based on 9 publications.

https://www.resque.info/includes/demo_profile.html

Independent Reuse

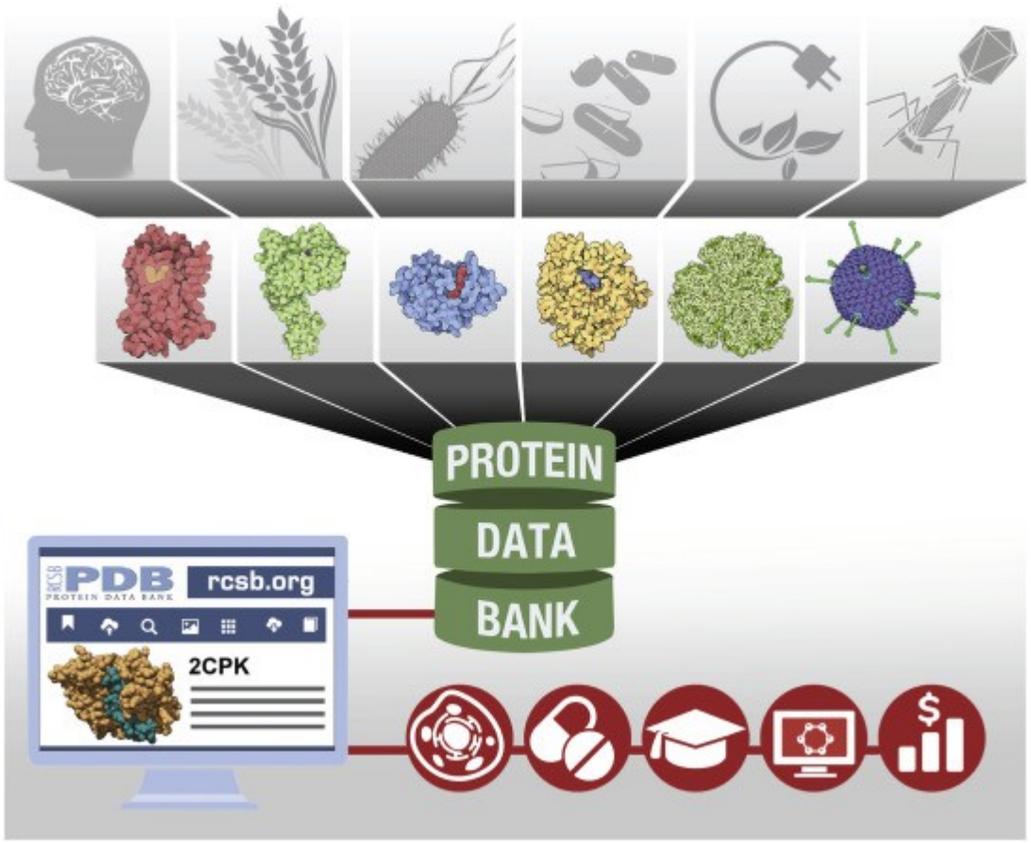
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Protein Data Bank (PDB): Fifty-three years young and having a transformative impact on science and society

Published online by Cambridge University Press: **20 February 2025**

Helen M. Berman and Stephen K. Burley

Show author details



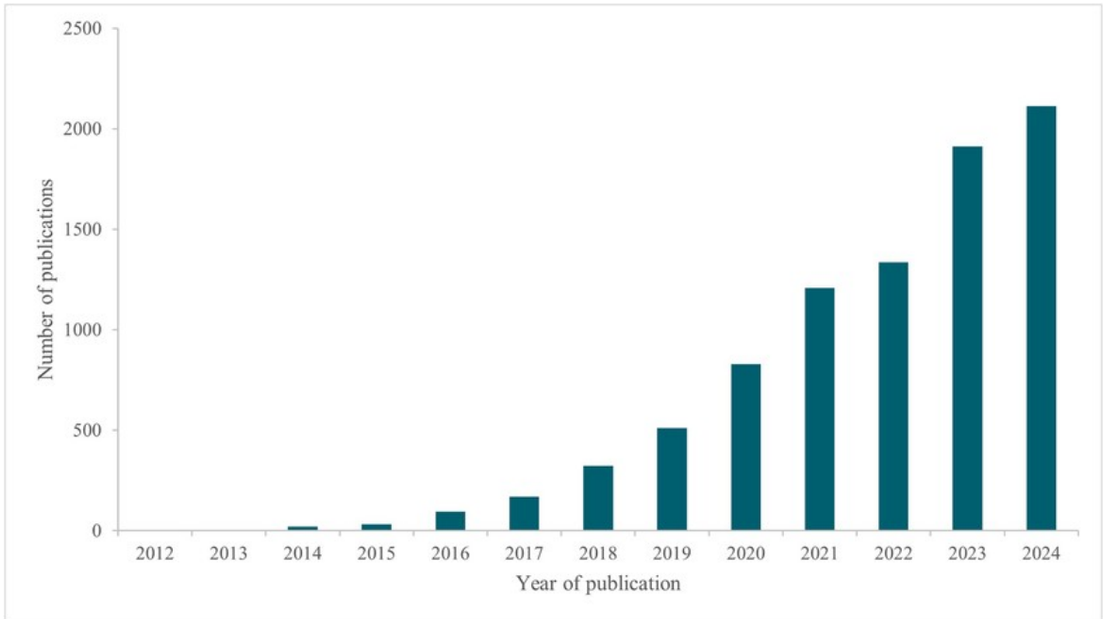
INVITED REVIEW | Open Access

biobank

UK Biobank: Transforming drug discovery and precision medicine

Jelena Bešević, Saredo Said, Reka Nagy, Yalda Jamshidi, Christopher D. Whelan, Lauren Carson, Martin K. Rutter, Adam J. Lewandowski, Mark Effingham ... See all authors

First published: 20 November 2025 | https://doi.org/10.1111/bph.70254 |



"Protein Data Bank archive helps…knowledge, benefiting basic and applied research and education."

Burley S **Impact of structural biologists and the Protein Data Bank on small-molecule drug discovery and development** Journal of Biological Chemistry, 2021; 296

Bešević, J., Said, S., Nagy, R., Jamshidi, Y., Whelan, C. D., Carson, L., Rutter, M. K., Lewandowski, A. J., Effingham, M., Collins, R., Lacey, B., & Allen, N. E. (2026). *UK Biobank: Transforming drug discovery and precision medicine*. *British Journal of Pharmacology*, 183(2), 234–248. https://doi.org/10.1111/bph.70254

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Thank you for your attention.